

Exercise 01: Selection Sort

The core step of the selection sort algorithm searches a subarray for the element with the minimum value, then swaps that element with the first element of the subarray — but only if their values differ. Write a C++ program named `'main01.cpp'` that satisfies the following requirements:

1. Include only the libraries `'iostream'`, `'string'`, `'cstdlib'`, `'ctime'`, and `'vector'`.
2. Define the `print()` function,

```
void print(vector<int>& q)
```

that displays the elements of `q` as a comma-separated list enclosed in square brackets.

3. Define the `moveMin()` function,

```
void moveMin(vector<int>& q, int i, int j)
```

that performs the core selection-sort step on the subarray `q` between `i` and `j` (inclusive), only if `i` and `j` are valid indices of `q`.

4. Testing in the `main()` function:
 - a. Initialize a vector object with random values.
 - b. Invoke `print()` to display the vector object.
 - c. Invoke `moveMin()` on the vector object with either user-provided or randomly generated indices.
 - d. Invoke `print()` again to display the vector object.
 - e. Repeat steps c and d with different indices.